

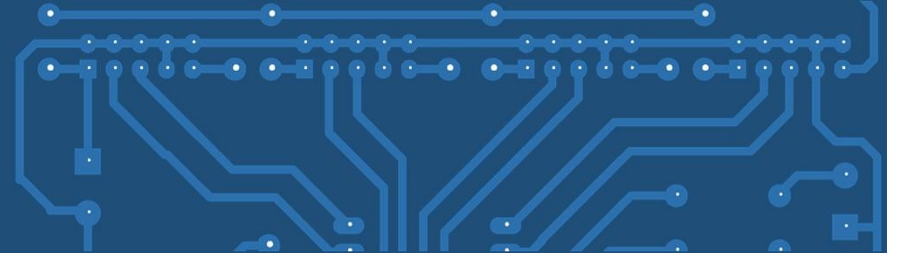
Topic 2 Reference Frames and Orientation

Part 2

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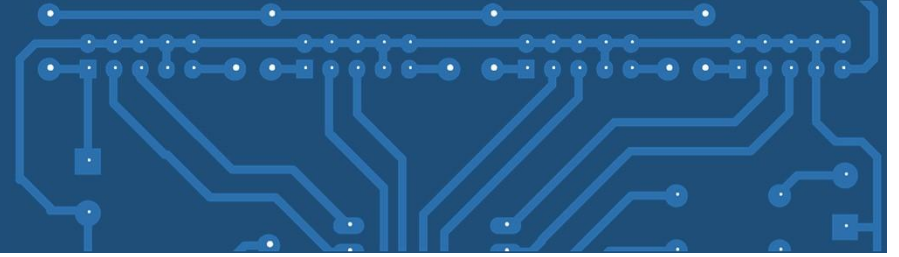
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Quaternions





Quaternions

$$[q_1, q_2, q_3, q_4]$$

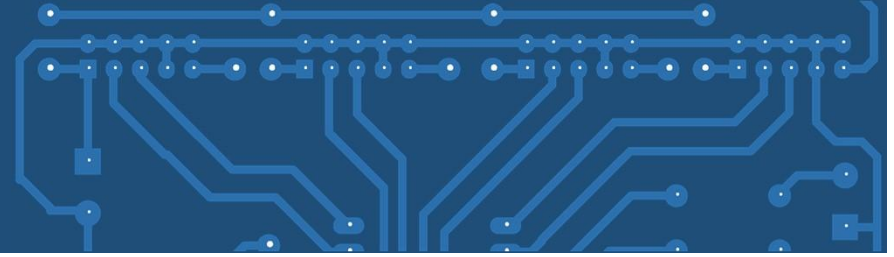
Advantages

- Singularity-free.
- Computationally efficient.

Disadvantages

- Non-intuitive.





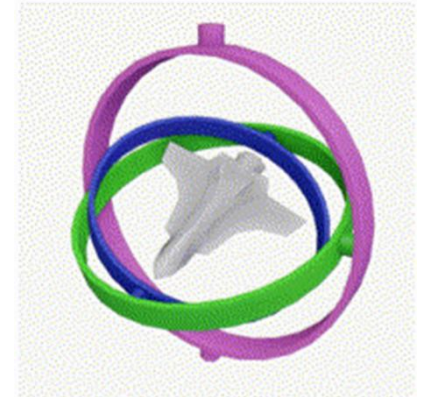
Why quaternions?

At pitch $\theta = \pm 90^\circ$, roll and yaw are coupled.

$\cos \theta = 0$ when $\theta = 90$ deg.

This can occur during extreme manoeuvres.

Unit quaternions avoid this Euler-angle singularity.



Quaternions can simplify attitude calculations.

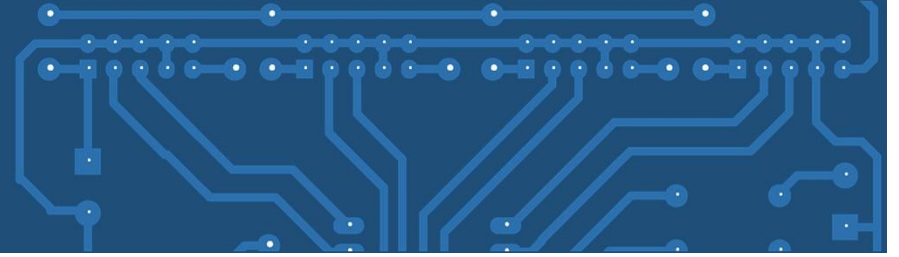
They support algebraic rotation calculations. Computational cost depends on the operation and implementation.

Gimbal Lock: <https://www.youtube.com/watch?v=zc8b2Jo7mno>



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What are quaternions?

Can also be thought of as a vector in four-dimensional space, $\mathbb{R}^4$

$$\mathbf{q} = [q_0 \quad q_1 \quad q_2 \quad q_3]$$

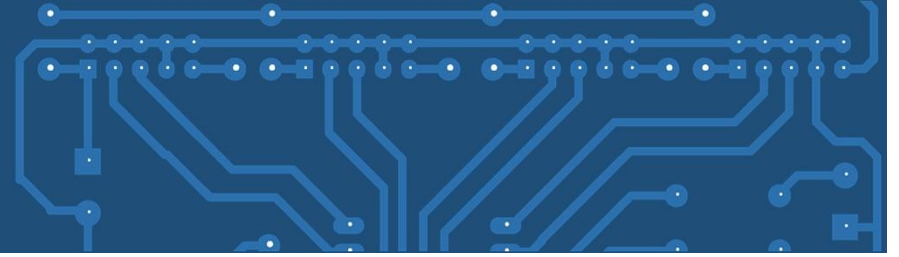
Also written as

$$\mathbf{q} = [s \quad \mathbf{v}]$$

where s is the scalar q_0 and $\mathbf{v}$ is the vector $[q_1 \quad q_2 \quad q_3]$

The product of non-scalar elements is non-commutative:

$$i^2 = j^2 = k^2 = ijk = -1$$



Quaternion meaning: axis and angle

A rotation has an axis and an angle θ about that axis.

A unit quaternion stores this using four numbers, scalar first.

$$q = [\cos(\theta/2), a \sin(\theta/2)]$$

a is a unit vector along the rotation axis.

The four components have squared sum 1.

q and $-q$ describe the same orientation.

Conjugate

$$q^* = q_0 - iq_1 - jq_2 - kq_3$$

Corollary

$$\begin{aligned}(q^*)^* &= q \\ (pq)^* &= q^*p^*\end{aligned}$$

Length (Norm)

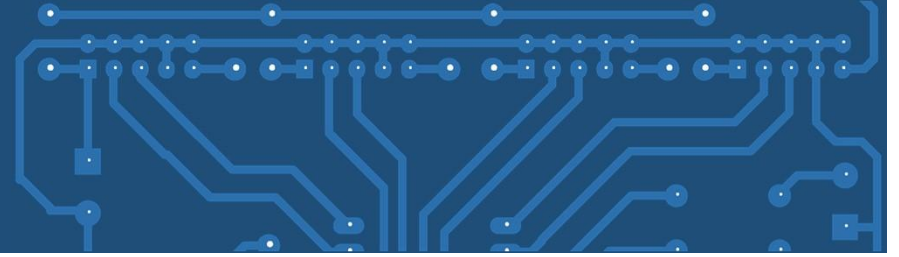
$$|q| = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}, \text{ equals } \mathbf{1} \text{ if it's a unit quaternion}$$

Normalization

$$q_{unit} = \frac{q}{|q|}$$

Inverse

$$q^{-1} = \frac{q^*}{|q|^2}$$



Example: normalisation and inverse

Given $q = [2, 0, 0, 2]$, calculate its norm.

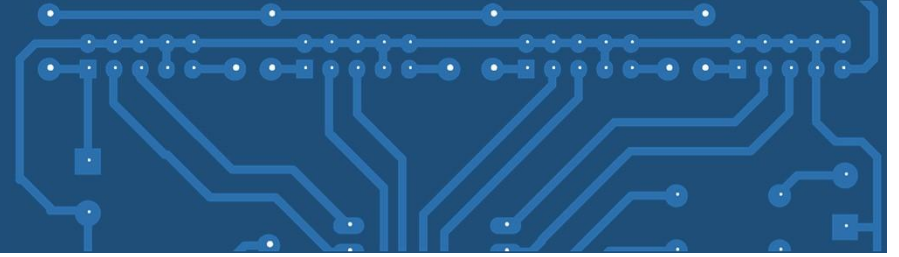
$$\|q\| = \sqrt{(2^2 + 0^2 + 0^2 + 2^2)} = \sqrt{8}$$

Divide every component by $\sqrt{8}$:

$$q_{\text{unit}} = [1/\sqrt{2}, 0, 0, 1/\sqrt{2}]$$

For this unit quaternion, the inverse is its conjugate:

$$q_{\text{unit}}^{-1} = q_{\text{unit}}^* = [1/\sqrt{2}, 0, 0, -1/\sqrt{2}]$$



Example: 90° rotation about z

Axis $a = [0, 0, 1]$ and half-angle $\theta/2 = 45^\circ$.

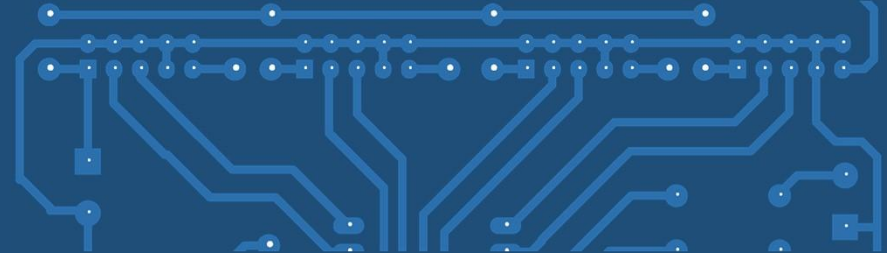
$$q = [\cos 45^\circ, 0, 0, \sin 45^\circ]$$

$$q = [1/\sqrt{2}, 0, 0, 1/\sqrt{2}]$$

Check: $(1/\sqrt{2})^2 + (1/\sqrt{2})^2 = 1$.

Your turn: construct q for a $+60^\circ$ rotation about x .

Use $\cos 30^\circ = \sqrt{3}/2$ and $\sin 30^\circ = 1/2$.



Quaternion operations

Rotation

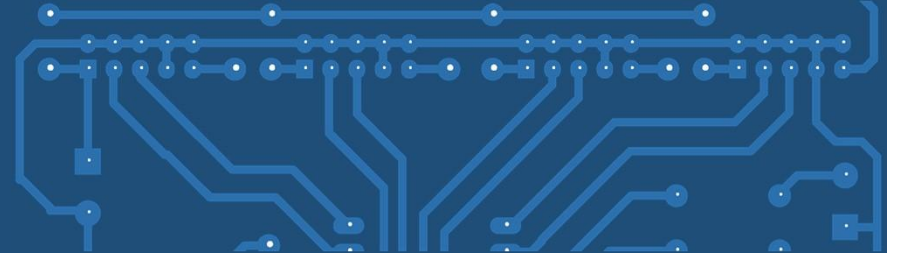
A quaternion can represent a rotation by angle θ around a unit vector $a = [a_x \ a_y \ a_z]$:

$$\mathbf{q} = \left[\cos \frac{\theta}{2} \quad a_x \sin \frac{\theta}{2} \quad a_y \sin \frac{\theta}{2} \quad a_z \sin \frac{\theta}{2} \right]$$

To rotate a point (vector) $p' = [p_x \ p_y \ p_z]$, convert it to a quaternion
 $p = 0 + v$

Then, the rotated vector is:

$$r = q \otimes p \otimes q^*$$



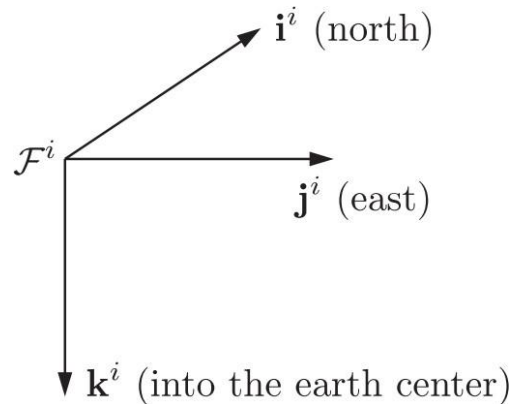
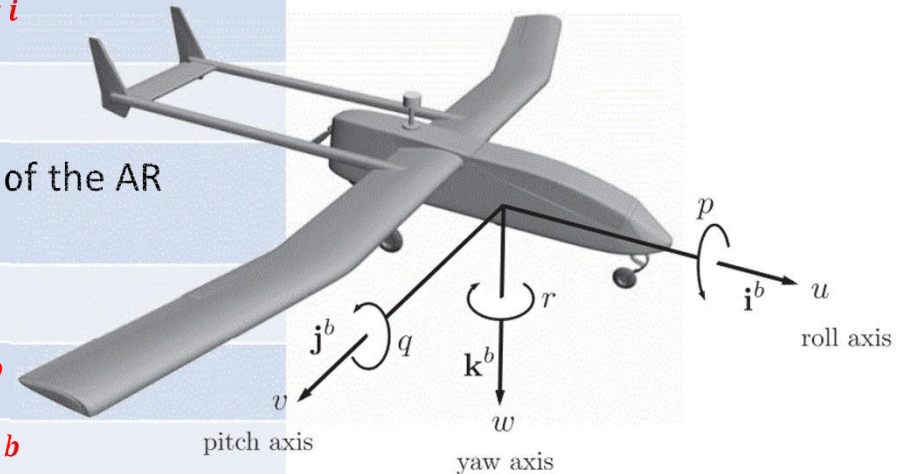
Kinematics



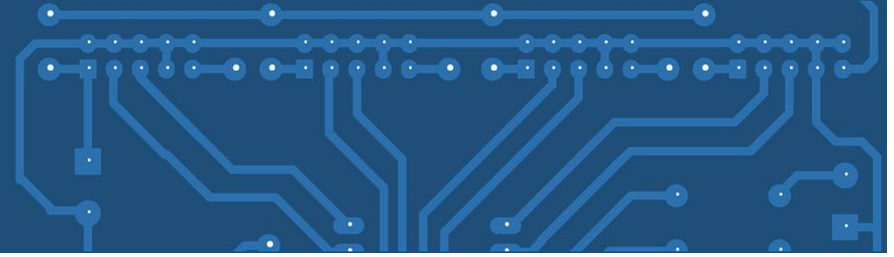
EEEE4138 Aerial Robotics

State variables definitions

Variable	Description
p_n	Inertial north position of the AR along $\mathbf{i}^i$ in $\mathcal{F}^i$
p_e	Inertial east position of the AR along $\mathbf{j}^i$ in $\mathcal{F}^i$
p_d	Inertial down position (negative of altitude) of the AR measured along $\mathbf{k}^i$ in $\mathcal{F}^i$
u	Body frame velocity measured along $\mathbf{i}^b$ in $\mathcal{F}^b$
v	Body frame velocity measured along $\mathbf{j}^b$ in $\mathcal{F}^b$
w	Body frame velocity measured along $\mathbf{k}^b$ in $\mathcal{F}^b$



Variable	Description
ϕ	Roll angle defined with respect to $\mathcal{F}^{v2}$
θ	Pitch angle defined with respect to $\mathcal{F}^{v1}$
ψ	Heading (yaw) angle defined with respect to $\mathcal{F}^v$
p	Roll rate measured along $\mathbf{i}^b$ in $\mathcal{F}^b$
q	Pitch rate measured along $\mathbf{j}^b$ in $\mathcal{F}^b$
r	Yaw rate measured along $\mathbf{k}^b$ in $\mathcal{F}^b$



Translational velocities

The components u , v , and w correspond to the inertial velocity of the vehicle projected onto the $\mathbf{i}^b$, $\mathbf{j}^b$, and $\mathbf{k}^b$ axes, respectively.

The translational position of the AR is usually measured and expressed in an inertial reference frame.

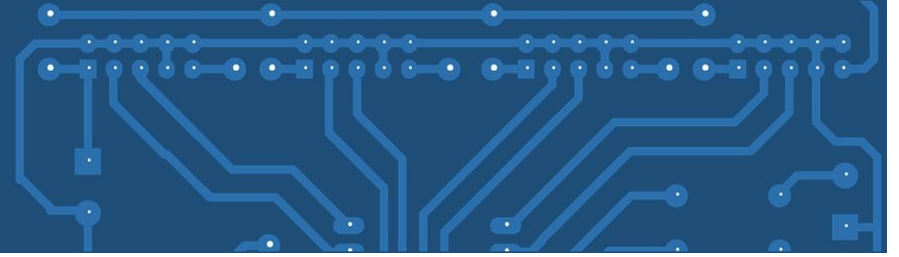
Relating the translational velocity and position requires differentiation and a rotational transformation;

$$\frac{d}{dt} \begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = R_b^v \begin{bmatrix} u \\ v \\ w \end{bmatrix} = (R_v^b)^T \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

Replacing for the rotation matrix yield

$$\begin{bmatrix} \dot{p}_n \\ \dot{p}_e \\ \dot{p}_d \end{bmatrix} = \begin{bmatrix} C\theta C\psi & S\phi S\theta C\psi - C\phi S\psi & C\phi S\theta C\psi + S\phi S\psi \\ C\theta S\psi & S\phi S\theta S\psi + C\phi C\psi & C\phi S\theta S\psi - S\phi C\psi \\ -S\theta & S\phi C\theta & C\phi C\theta \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$





Exercise: body velocity to NED

A level UAV points 30° east of north.

Body velocity is $[u, v, w] = [4, 0, 0]$ m/s.

Find its NED velocity and displacement after 5 s.

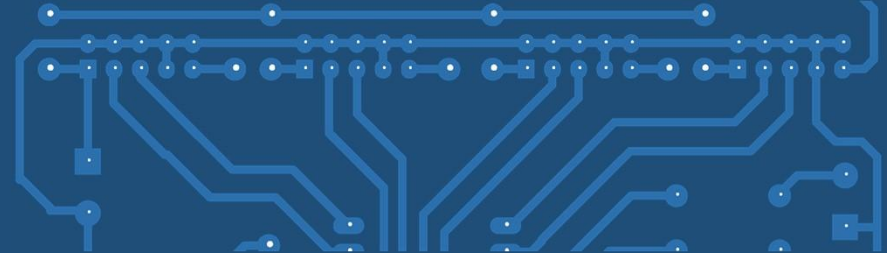
For level flight:

North rate $= u \cos\psi - v \sin\psi$

East rate $= u \sin\psi + v \cos\psi$; down rate $= w$

Assume constant heading and velocity.





Angular rates

The components p , q , and r correspond to the angular rates defined in the body frame $\mathcal{F}^b$.

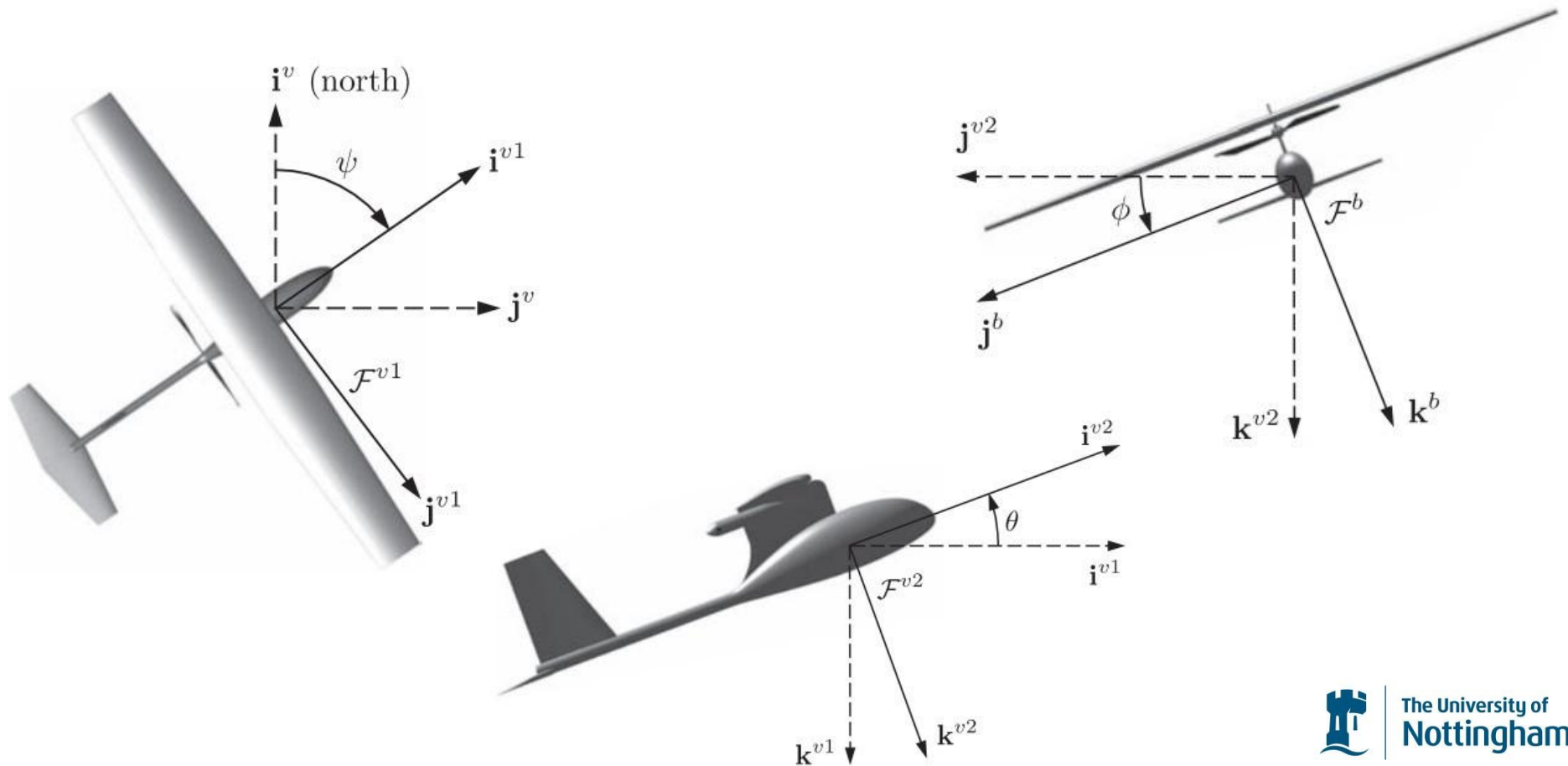
The angular positions (Euler angles) ϕ , θ , and Ψ are defined in three different frames:

- Roll angle ϕ is a rotation from $\mathcal{F}^{v2}$ to $\mathcal{F}^b$ about the $\mathbf{i}^{v2}=\mathbf{i}^b$ axis;
- Pitch angle θ is a rotation from $\mathcal{F}^{v1}$ to $\mathcal{F}^{v2}$ about the $\mathbf{j}^{v1}=\mathbf{j}^{v2}$ axis; and
- Yaw angle Ψ is a rotation from $\mathcal{F}^v$ to $\mathcal{F}^{v1}$ about the $\mathbf{k}^v=\mathbf{k}^{v1}$ axis.

The body-frame angular rates can be expressed in terms of the derivatives of the Euler angles, provided that the proper rotational transformations are carried out.

Angular rates

- Roll angle ϕ is a rotation from $\mathcal{F}^{v2}$ to $\mathcal{F}^b$ about the $\mathbf{i}^{v2}=\mathbf{i}^b$ axis;
- Pitch angle θ is a rotation from $\mathcal{F}^{v1}$ to $\mathcal{F}^{v2}$ about the $\mathbf{j}^{v1}=\mathbf{j}^{v2}$ axis; and
- Yaw angle Ψ is a rotation from $\mathcal{F}^v$ to $\mathcal{F}^{v1}$ about the $\mathbf{k}^v=\mathbf{k}^{v1}$ axis.



Angular rates

- Roll angle ϕ is a rotation from $\mathcal{F}^{v2}$ to $\mathcal{F}^b$ about the $\mathbf{i}^{v2}=\mathbf{i}^b$ axis;
- Pitch angle θ is a rotation from $\mathcal{F}^{v1}$ to $\mathcal{F}^{v2}$ about the $\mathbf{j}^{v1}=\mathbf{j}^{v2}$ axis; and
- Yaw angle Ψ is a rotation from $\mathcal{F}^v$ to $\mathcal{F}^{v1}$ about the $\mathbf{k}^v=\mathbf{k}^{v1}$ axis.

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + R_{v2}^b(\phi) \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + R_{v2}^b(\phi) R_{v1}^{v2}(\theta) \begin{bmatrix} 0 \\ 0 \\ \dot{\Psi} \end{bmatrix}$$

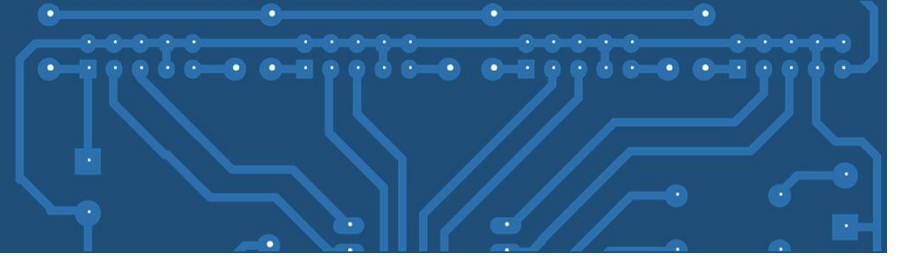
$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\phi & S\phi \\ 0 & -S\phi & C\phi \end{bmatrix} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & C\phi & S\phi \\ 0 & -S\phi & C\phi \end{bmatrix} \begin{bmatrix} C\theta & 0 & -S\theta \\ 0 & 1 & 0 \\ S\theta & 0 & C\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\Psi} \end{bmatrix}$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -S\theta \\ 0 & C\phi & S\phi C\theta \\ 0 & -S\phi & C\phi C\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\Psi} \end{bmatrix}$$

Inverting this gives the Euler-angle rate mapping:

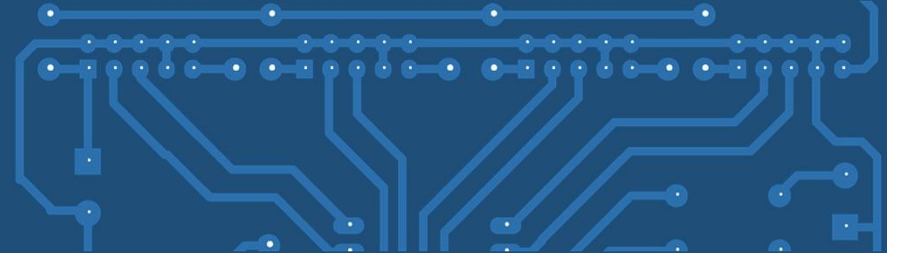
$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\Psi} \end{bmatrix} = \begin{bmatrix} 1 & S\phi T\theta & C\phi T\theta \\ 0 & C\phi & -S\phi \\ 0 & S\phi/C\theta & C\phi/C\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$





Homogeneous Transformation



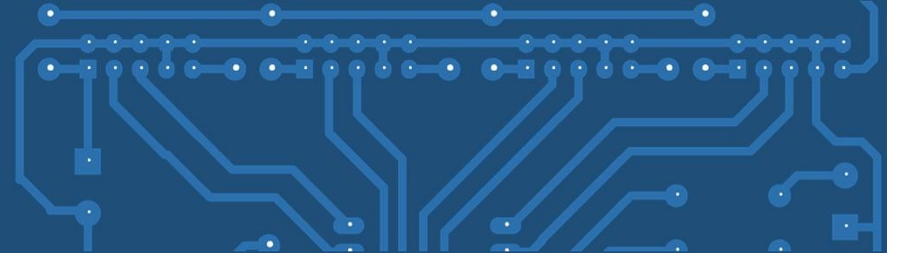


Homogeneous Transformation...

Homogeneous transformations combine rotations and displacements into a single transformation matrix;

$$H = \left[\begin{array}{c|c} R & D \\ \hline 0 & 1 \end{array} \right]$$

$$H = \left[\begin{array}{ccc|c} n_x & o_x & s_x & d_x \\ n_y & o_y & s_y & d_y \\ n_z & o_z & s_z & d_z \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$



Homogeneous Vectors...

For points, append 1 as shown below. For directions, append 0 so translation has no effect.

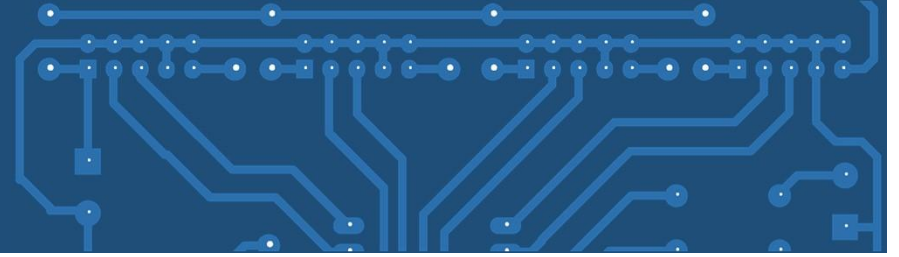
$$P = \begin{bmatrix} p \\ 1 \end{bmatrix}$$

$$P^0 = H_1^0 P^1$$

$$P = \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix}$$

$$P^0 = \begin{bmatrix} n_x & o_x & s_x & d_x \\ n_y & o_y & s_y & d_y \\ n_z & o_z & s_z & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x^1 \\ p_y^1 \\ p_z^1 \\ 1 \end{bmatrix}$$





Exercise: locating a camera

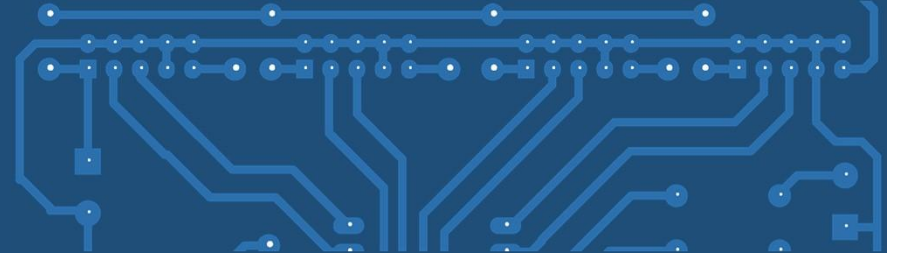
A level UAV faces east. Its NED position is $[10, 20, -5]$ m.

The camera is 0.30 m forward and 0.10 m body-down.

Find the camera position in NED coordinates.

1. Rotate the body offset $[0.30, 0, 0.10]$ into NED.
2. Add the UAV position.

Hint: body forward points east; positive down is below.



Inverse Transformation...

Homogeneous transformations can be reversed by inverting H . The following shortcut can be used to do that:

$$H = \begin{bmatrix} R & D \\ 0 & 1 \end{bmatrix}$$

$$H^{-1} = \begin{bmatrix} R^T & -R^T D \\ 0 & 1 \end{bmatrix}$$